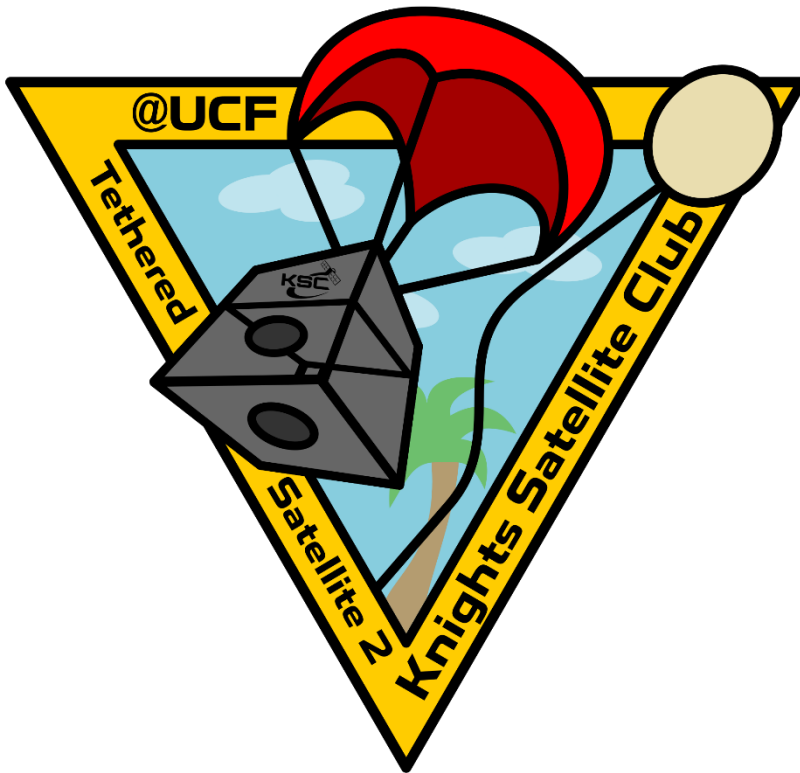




Critical Design Review (CDR)

Tethered Satellite 2 Mission A

Knight's Satellite Club



Introduction

Motivation

The Knight's Satellite Club (KSC), a registered student organization at the University of Central Florida. Introducing a second iteration to their tethered satellite project. With the Tethered-Satellite also referred to as T-Sat being one of the many projects founded, designed and engineered by members of the Knight's Satellite Club (KSC). Through this project and its multiple iterations, it has allowed students of all different types of experiences to come together to learn and create Cube Satellites.

Tethered-Satellite 2 Mission A (T-Sat 2A) underwent significant changes throughout its development. The final design incorporated knowledge gained from previous project leads, members, testing, and the practical constraints encountered during fabrication and integration. As a result, the completed system differs in several areas from the concepts originally presented in the Preliminary Design Review (PDR), Mission Concept Review (MCR), and T-Sat 2A budget documentation. The previous documentation established the initial design intent and requirements, while this document focuses on the development of the system that was ultimately fabricated, tested, and flown.

The purpose of this document is therefore not to replace or undermine the work presented in the previous design documentation. Instead, it serves as a final technical record of the T-Sat 2A mission and the engineering process used to achieve its objectives. The final design retained several of the fundamental concepts established during the earlier design phases, including the use of a 2U CubeSat architecture and an internal separation system. However, the implementation of these concepts evolved substantially as the team progressed from preliminary design into fabrication, integration, testing, and flight operations.

Where applicable, this document identifies the differences between the original design intent and the final implementation, explains the engineering reasoning behind those changes, and documents problems encountered throughout the development process. Particular attention is given to design iterations, manufacturing and integration challenges, testing results, failures or unexpected behavior, and the solutions implemented by the team.

The goal of this approach is to provide more than a description of the completed CubeSat. By documenting both successful and unsuccessful design decisions, this CDR is intended to serve as a technical reference for future KSC members. The information presented should allow future teams to understand how the final system was developed, why specific design decisions were made, what lessons were learned during the mission, and, where appropriate, how the design and associated processes could be reproduced or improved in future iterations.

The original and subsequent project documentation, along with applicable open-source code and design files, are maintained through the KSC Wiki and the respective project repositories. These resources provide additional technical information and supporting files for the systems discussed throughout this document ([FrancescaAlfaroGIT/SethJonesGIT](#)) and [KSC WIKI](#).

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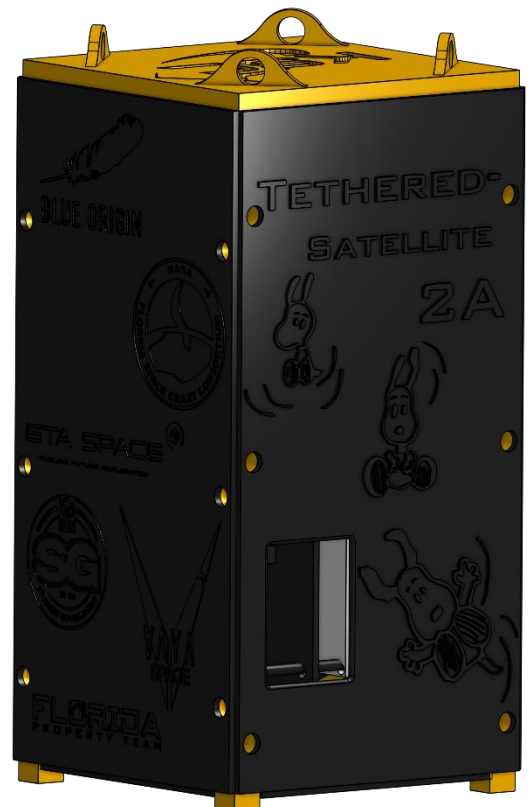
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System Overview

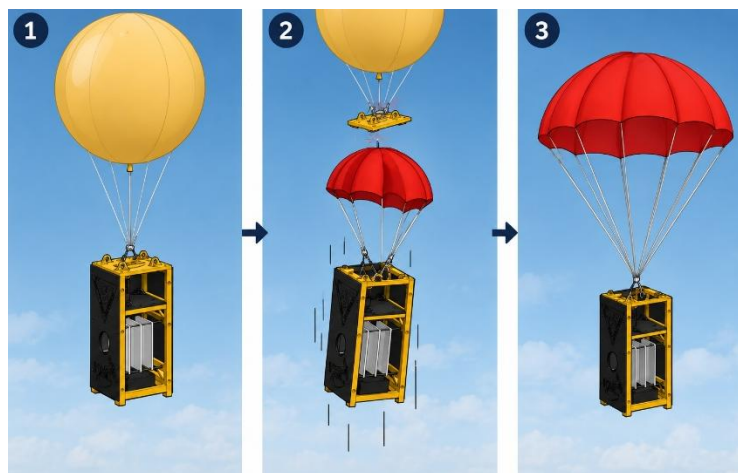
Tethered-Satellite 2 Mission A (T-Sat 2A) represented the third iteration of the Tethered-Satellite project and introduced a significant expansion in both the CubeSat architecture and mission capabilities compared with previous iterations. While maintaining the fundamental requirement of operating as a tethered satellite, T-Sat 2A was designed around a 2U CubeSat architecture, providing approximately twice the internal volume of previous designs and allowing the team to incorporate additional cameras, sensors, electronics, and a more capable deployment system.

The mission was designed to operate at an altitude of approximately 500 ft while remaining physically connected to the ground through a tether (based off [FFA Regulations](#)). A 600 g weather balloon filled with helium served as the lifting system. These operational constraints established the basic architecture of the mission while allowing the team to investigate new approaches to deployment, imaging, sensing, and CubeSat design.

During the early stages of development, several concepts were considered for the descent and recovery system. One of the more ambitious concepts involved developing a self-controlling parafoil capable of autonomous or manual flight control during descent. Although this concept was ultimately removed from the T-Sat 2A design because of the project's one-semester development timeline, the concept remains a potential area for future KSC projects. The final mission retained a dedicated deployment system while simplifying the descent system to better accommodate the available development time and resources.

The final descent system consisted of a 36-inch nylon parachute and a servo-actuated release mechanism. The servo was controlled remotely from the ground station and rotated approximately 90° to actuate the release mechanism by cutting a fishing-wire connection. This configuration allowed the launch team to initiate deployment while on site rather than relying solely on an automatic release sequence. The deployment architecture also provided a physical connection between the upper panel and the 2U CubeSat prior to release.

Figure 1: Demonstration of Separation System with CubeSat

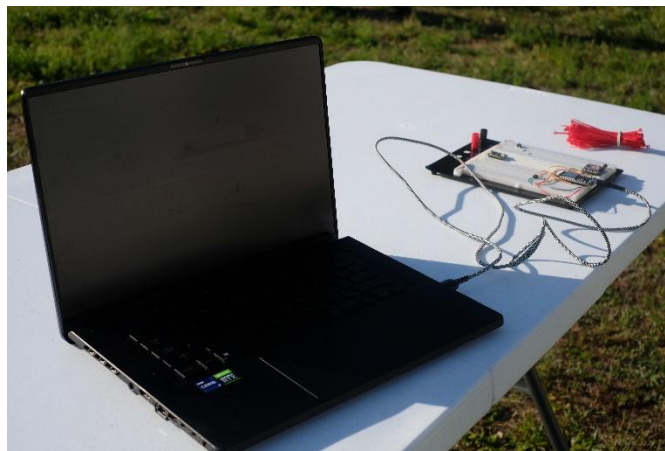


The increased internal volume of the 2U configuration also allowed T-Sat 2A to expand its imaging and sensing capabilities. The CubeSat incorporated two 5 MP cameras to provide

multiple fields of view along with an altimeter and other onboard sensors. These components were integrated with an ESP32-based control system, SD card storage, an RF LoRa communication module, and the servo used for deployment. The final electrical architecture was designed to reuse applicable components and modules from previous T-Sat iterations where practical while introducing new printed circuit boards (PCBs) and software required to support the expanded system.

The ground station provided the primary interface between the launch team and the CubeSat. It incorporated an ESP32, an RF LoRa module, and a user-operated button. When activated by the launch team, the ground station transmitted the deployment command to the CubeSat, allowing the onboard servo to actuate the release mechanism. This provided the team with direct control over the deployment sequence during mission operations.

Figure 2: T-Sat 2A Ground Station Set-Up



To support the expanded electrical architecture, the CubeSat was divided between two PCBs that operated together through JST crimp connections. The boards distributed power and signals among the onboard modules while maintaining a modular electrical architecture. T-Sat 2A also utilized two separate lithium-ion batteries: a 500-mA battery dedicated to the servo system and a 420-mA battery supporting the remaining electronics and cameras. The reasoning behind this power architecture, including the advantages and limitations identified during development, is discussed in greater detail in the Electrical System section.

The CubeSat housing was manufactured primarily through 3D printing using PETG. The housing was designed with tight tolerances to allow the structure to be assembled and disassembled during testing while minimizing unnecessary hardware. Side panels were secured to the main housing using M5 screws, while M3 screws were primarily used for housing connection points, servo mounting, and the bottom panel. Heated inserts were incorporated into the 3D-printed structure to provide more durable threaded connection points and maintain structural rigidity.

The two PCBs were mounted using standoffs connected to heated inserts within the CubeSat housing. This arrangement provided sufficient structural support while allowing the boards to be removed and repositioned during testing and integration. The two batteries were similarly mounted adjacent to the PCBs using standoffs, with electrical tape used to limit battery movement and reduce the possibility of movement affecting the JST connections.

The final T-Sat 2A system therefore consisted of several integrated subsystems: the mechanical CubeSat structure, electrical and power systems, onboard computing and data storage, imaging and sensing systems, RF communication system, ground station, and parachute deployment system. Each subsystem is discussed in greater technical detail in the sections that follow, including its design process, implementation, testing, problems encountered, and final configuration.

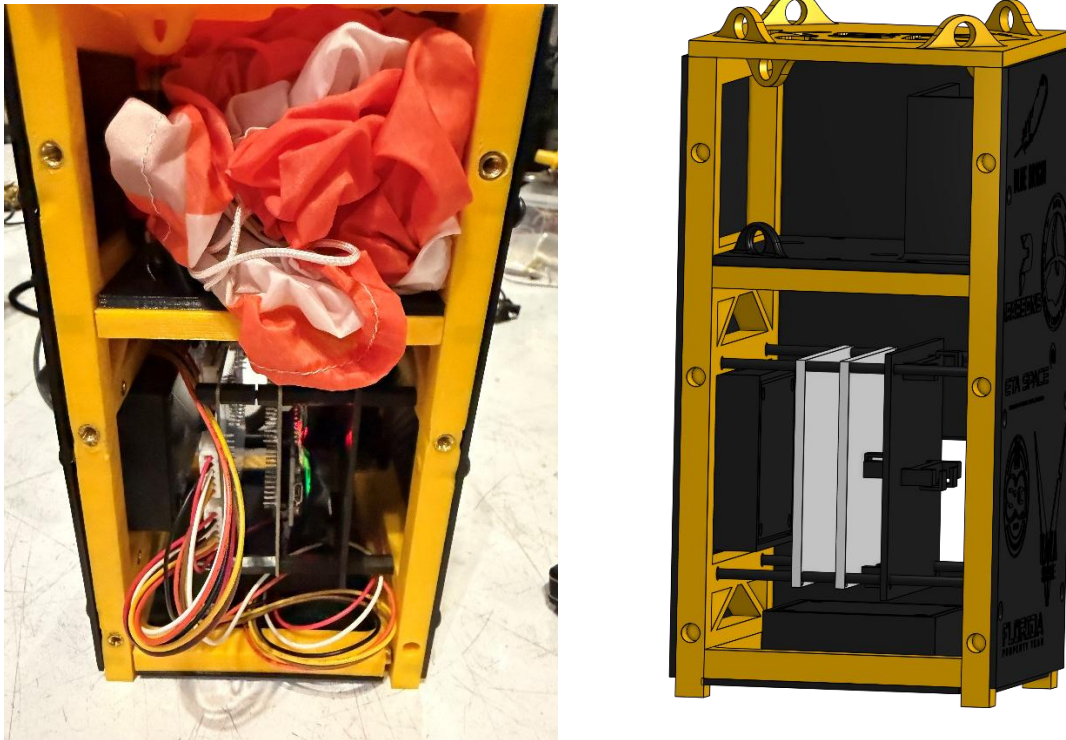


Figure 3: Inside View of a Launch Ready T-Sat 2A

Ultimately, the objective of T-Sat 2A extended beyond the successful operation of a single CubeSat-scale system. The project provided an opportunity for students to experience the complete engineering development process, including preliminary design, hardware development, manufacturing, electrical integration, software development, testing, troubleshooting, and mission operations. By documenting the final configuration as well as the design iterations and problems encountered during development, this CDR is intended to preserve the engineering knowledge gained from T-Sat 2A and provide a technical reference for future KSC teams developing subsequent T-Sat missions.

Mechanical Design and Development

The development of T-Sat 2A as KSC's first 2U T-Sat configuration introduced several mechanical design considerations that were not present to the same extent in previous iterations. The primary concerns were maintaining the required structural integrity while controlling mass, providing sufficient space for the electrical and payload systems, and creating a structure that could be manufactured and iterated quickly. Because the project was developed within a limited timeline, the mechanical design was also intended to allow components to be reproduced or modified without requiring significant changes to the overall manufacturing process.



Structural Architecture

The main CubeSat housing was designed around a modular and repeatable geometry. The two primary housing sections that contained the heated threaded inserts for structural connections were designed as the same part, with one section duplicated and reversed during assembly. This approach reduced the number of unique components that needed to be designed and manufactured. Consequently, modifications to the housing could be performed on a single CAD part and then incorporated into both sides of the final structure.

Several structural features were incorporated into the housing to increase rigidity while limiting unnecessary material. Triangular wall features were added to provide additional

structural support and reduce the likelihood of deformation along the primary linear sections of the housing. These features also supported the horizontal structural members used to mount the internal electronics.

Open regions were intentionally retained throughout portions of the housing. Rather than completely filling the structure with material, these openings reduced overall mass and filament consumption while maintaining the structural members required to support the CubeSat. This approach was particularly important because the CubeSat was lifted by a weather balloon, making the mass of the complete system a significant design consideration.

The horizontal structural members incorporated heated inserts that provided mounting locations for the standoffs supporting the two PCBs and battery mounting hardware. This allowed the internal electronics to be mechanically supported without requiring the PCB mounting system to be directly integrated into the external housing geometry.

Internal Component Mounting

The internal mounting system was designed around standoffs connected to heated inserts within the 3D-printed housing. This configuration provided a rigid mounting interface for the PCBs while allowing the electronics to be removed and repositioned during testing and integration.

The same general mounting philosophy was applied to the battery system. The two lithium-ion batteries were positioned adjacent to the PCBs and mechanically constrained using standoff-based mounting features. Electrical tape was additionally used to limit battery movement and reduce the possibility of movement affecting the JST connections between the batteries and PCBs.

This mounting method provided a balance between structural support, accessibility, and ease of integration. Because the CubeSat required repeated testing and modifications during development, the ability to access and reposition internal components was considered more useful than permanently encapsulating the electronics within the structure.

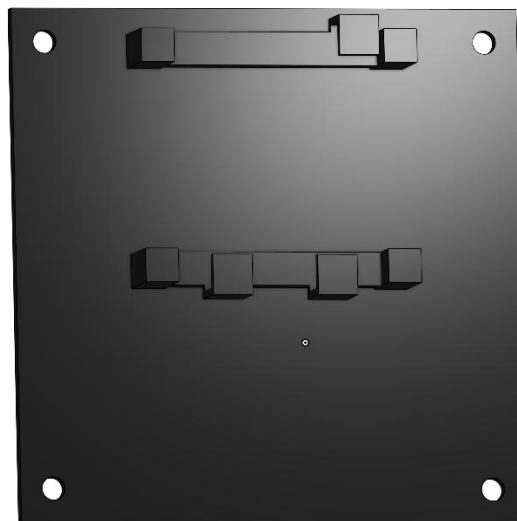
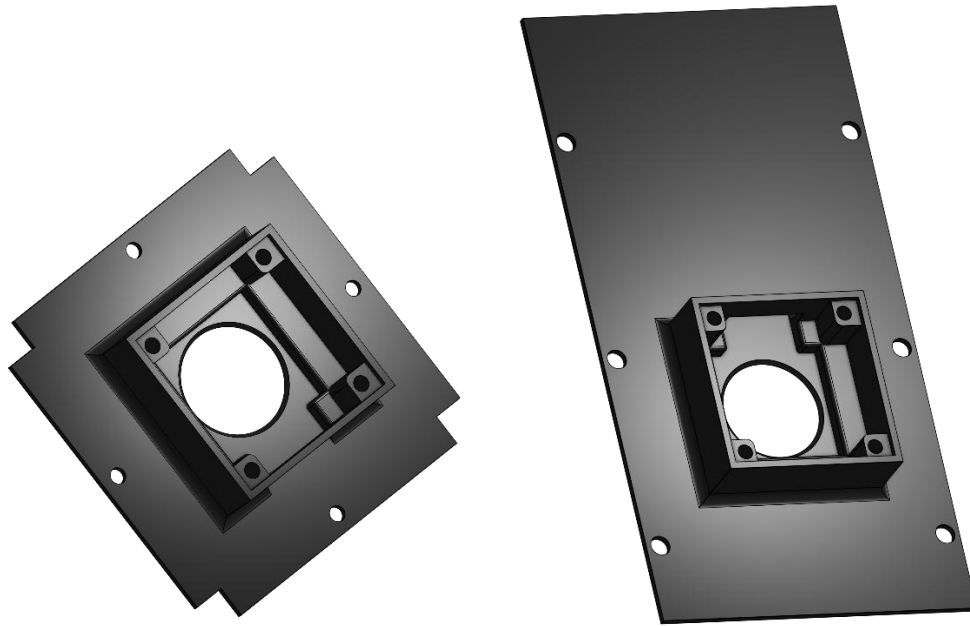


Figure 4: T-Sat 2A Battery Holder

Figure 5: T-Sat 2A Inside View of Camera Holder



Camera Mounting

The camera mounting system was based on the camera-holder design used during the previous T-Sat iteration. Although the previous design had experienced a failure during drop testing, the camera holder itself remained intact. The existing design was therefore considered sufficiently robust to serve as a starting point for T-Sat 2A rather than requiring a completely new camera mounting architecture.

The primary modification was the addition of larger fillets at the connection points between the camera holder and the CubeSat panel. Increasing the fillet geometry provided a larger transition between the mounting features and the panel while reducing the concentration of material around the connection region. Reusing and improving the previous design also allowed the mechanical team to dedicate development time to other areas of the T-Sat 2A mechanical system.

Top Panel and Deployment Interface

The structural trusses and horizontal members maintained the established mounting geometry at the upper connection points. An opening was incorporated into the upper portion of the structure to accommodate the approximately 1.5-inch top base plate. The base plate was retained through the interaction between the designed tolerances and mechanical fasteners, with heated inserts and M3 bolts providing the primary threaded connections.

The top base plate supported the additional panel associated with the deployment mechanism. This panel housed the servo and provided the connection points for the fishing-line retention system that temporarily connected the CubeSat body to the upper balloon assembly.

The servo panel was designed with a dedicated clearance region around the servo actuator. A cutting blade was attached to the end of the servo mechanism and rotated

approximately 90° during deployment. The clearance around the actuator was designed to prevent the blade from unintentionally contacting the parachute, housing, or other components during operation.

Fishing-Line Retention and Release System

The connection between the CubeSat body and the upper panel was implemented using fishing line routed through four connection points. During the initial testing of the line-routing system, it was discovered that the geometry surrounding the servo created interference with the intended trajectory of the fishing line. The cutout originally provided around the servo prevented the line from following a sufficiently direct path between the opposing connection points.

To correct this issue, an arc-shaped cutout was added to the affected region of the panel. The revised geometry allowed the fishing line to follow a more direct path while maintaining clearance from the servo blade. This was an example of an integration issue that was not fully apparent during the initial CAD development but became evident during physical testing of the completed mechanism.

The four-point connection system was designed to distribute the retention load between the upper panel and the CubeSat housing. The fishing line was secured using a Uni Knot/Duncan Loop configuration. Before proceeding with the design, the mechanical team performed a load test to determine whether the connection points and fishing-line configuration could withstand the expected downward loading.

A force gauge previously used during T-Sat balloon-launch testing was used to apply a downward load to the connection system. The test was performed without actuating the cutting blade so that the strength of the connection itself could be evaluated independently of the release mechanism. The connection remained intact beyond the maximum load measurable by the available gauge. Although this test did not establish the exact ultimate failure load because the applied load exceeded the measurement range of the instrument, it provided sufficient evidence for the team to continue with the four-point retention concept.

One of the primary mechanical risks identified with this system was the proximity between the fishing line and the servo-mounted cutting blade. The blade was positioned only a few millimeters from the line so that a 90° servo rotation would reliably contact and cut the line. However, this small separation also introduced the possibility of premature deployment if the blade were accidentally actuated or if the line shifted into the blade's path. The mechanical team therefore identified unintended blade activation or premature line cutting as a significant risk associated with the final release mechanism.

Parachute Integration and Drop Testing

After the final housing geometry was established, a dedicated dummy housing was manufactured for parachute deployment testing. The dummy housing incorporated a screw-mounted hook at the center of the structure, providing a location to attach the parachute without requiring the flight electronics or complete flight hardware to be installed.

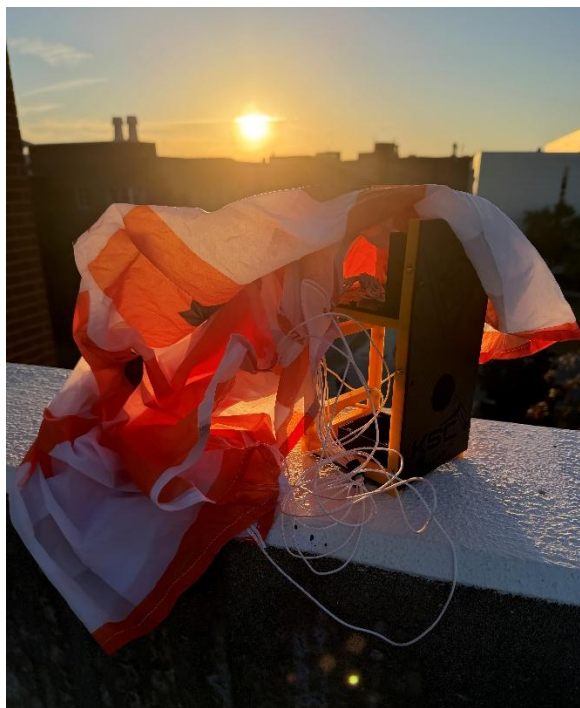
Initial drop testing was conducted from the top of the UCF parking garage at an approximate height of 50 ft. Because this height was substantially lower than the intended

mission altitude, the tests were treated primarily as deployment and mechanical-behavior tests rather than full-scale flight simulations.

Multiple drops were performed with the parachute installed. Members of the team positioned a tablecloth at ground level to catch the test article and reduce the risk of damage while observing the parachute deployment sequence. The primary objective was to determine whether the parachute could exit the approximately 0.5-inch-deep upper CubeSat section and begin deploying as intended.

Although the available test height limited the amount of time available for the parachute to fully inflate, slow-motion video from the tests showed that a significant portion of the parachute was able to exit and begin deployment. The results suggested that the deployment mechanism and available packing volume were capable of initiating parachute deployment, while also demonstrating the limitations of conducting parachute testing from a relatively low altitude.

Figure 6: T-Sat 2A Dropping Test Set-Up



4.7 Material Selection

PETG was selected as the primary material for the T-Sat 2A housing based on its durability demonstrated during previous T-Sat development, its suitability for the intended manufacturing process, and its accessibility to the team. PETG also provided a practical balance between mechanical performance, manufacturing speed, cost, and ease of iteration.

Alternative materials, including nylon and carbon-fiber-reinforced materials, were considered as potential options for reducing mass or increasing mechanical performance. However, the expected benefit of these materials was not considered sufficient to justify the additional manufacturing complexity and cost for the requirements of T-Sat 2A. The use of

PETG also allowed the team to rapidly reproduce and modify components as problems were identified during testing.

Side Panels and Environmental Enclosure

The CubeSat side panels served both structural and design purposes. The panels provided the external enclosure for the CubeSat while also providing surfaces for KSC, T-Sat, and other project-related graphics and logos.

Beyond their visual purpose, the panels helped create a more enclosed internal environment for the CubeSat's electronics. Maintaining the electronics within the CubeSat housing provided additional protection from the external environment during launch and flight operations.

The side panels were primarily secured to the housing using M5 screws. Heated inserts were incorporated into the 3D-printed structure to provide reusable threaded connection points and improve the durability of the attachment locations. M3 hardware was primarily used for smaller mounting applications, including housing connection points, servo mounting, and bottom-panel attachment.



Figure 7: T-Sat 2A All Created Side Panels

CAD and Manufacturing Process

All major mechanical components were modeled and developed using Onshape. The use of a cloud-based CAD environment allowed multiple members of the mechanical team to work within the same design environment and observe updates as development progressed.

This collaborative approach was particularly useful during integration because mechanical members could see how their components interacted with the electronics, payload, and other structural elements being developed by the rest of the team. It also allowed design changes to be communicated more efficiently and reduced the likelihood of individual team members working from outdated versions of a component.

The combination of a repeatable housing architecture, 3D printing, standardized mounting hardware, and collaborative CAD allowed the mechanical system to be rapidly iterated

throughout development. Components could be modified, reprinted, assembled, and tested without requiring extensive changes to the remainder of the CubeSat.

Final Mechanical Design

The final T-Sat 2A mechanical system consisted of a PETG 3D-printed 2U housing, removable side panels, internal structural members, PCB and battery mounting hardware, camera mounts, a dedicated servo panel, and a four-point fishing-line retention and release system. The structure was designed to balance mass, rigidity, manufacturability, accessibility, and integration requirements.

The final mechanical configuration was the result of multiple design iterations rather than a single completed design. Several features, including the camera mounts, fishing-line routing, servo clearance, and parachute deployment interface, were refined through physical testing and integration. These iterations demonstrated the importance of evaluating mechanical systems not only through CAD but also through physical assembly and testing, where interference, fitment, and operational issues could be identified that were not immediately apparent in the digital model.

Electrical and Avionics Development

Electrical Architecture and Component Selection

The electrical and avionics system for T-Sat 2A was developed to support four primary mission functions: image acquisition, altitude measurement, wireless communication, and parachute deployment. The increased internal volume of the 2U configuration allowed the team to incorporate two cameras, additional sensing capabilities, and a more modular electrical architecture while building upon components and experience from previous T-Sat iterations.

The primary flight computer was an ESP32 microcontroller, which coordinated the onboard peripherals through the flight software. The system incorporated two Arducam cameras, a BMP388 barometric altimeter, an SD card module for local data storage, an RFM95W LoRa radio module for communication, and a servo motor for the parachute release mechanism.

Component selection was influenced by the team's previous experience, available development time, and the objective of reusing applicable hardware where possible. An accelerometer was initially considered for the avionics system. However, recurring software issues experienced with the accelerometer during the T-Sat 0 and T-Sat 1 projects contributed to the decision to remove it from the final design. Eliminating the component reduced the software and integration requirements within the available development timeline.

The BMP388 was selected as the primary altitude sensor. Although the device also supports temperature measurement, temperature data was not a primary mission requirement and was not prioritized in the final software implementation.

The final electrical architecture was divided between two custom-designed printed circuit boards (PCBs). This configuration provided a means of organizing the increased number of electrical connections required by the two-camera system while maintaining compatibility with the internal mechanical structure.

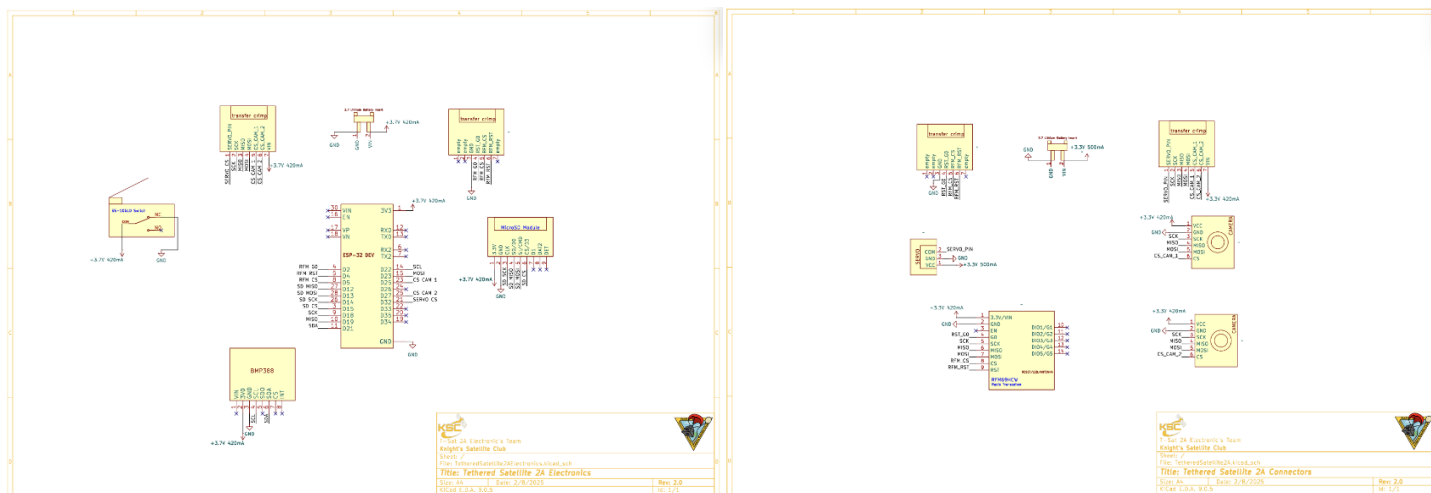


Figure 8: T-Sat 2A Both PCB Schematic

PCB Design and Development

PCB development initially began in EasyEDA because of its collaborative design capabilities. During the design process, the team encountered difficulties involving component symbols, footprints, and electrical connections. These challenges affected the PCB development workflow and created a need for greater control over the board design and component libraries.

To address these issues, the PCB design was transferred to KiCad. The new development environment allowed the team to create and modify custom footprints and symbols for components that were not adequately represented by the available libraries. The completed PCB designs were subsequently manufactured through JLCPCB.

The final system incorporated two custom PCBs, each measuring approximately 71 mm × 71 mm. The boards included mounting holes, through-hole components where applicable, battery connections, and JST connectors for interfacing with the electrical subsystems. Standardized connectors supported modular assembly and allowed individual components to be disconnected during troubleshooting and testing.

The use of two PCBs also supported the mechanical constraints of the 2U housing. The boards were designed to be mounted using the internal standoffs and mounting features described in the mechanical design section. This allowed the electrical system to be integrated with the housing while maintaining access to the boards during assembly and testing.

During manufacturing and assembly, one surface-mount battery connector was installed in an orientation opposite to the intended design. The error did not prevent the completed system from being used, but it represented a manufacturing or assembly issue that should be corrected in future PCB revisions. This experience emphasized the importance of reviewing component orientation and connector placement before manufacturing and during board assembly.

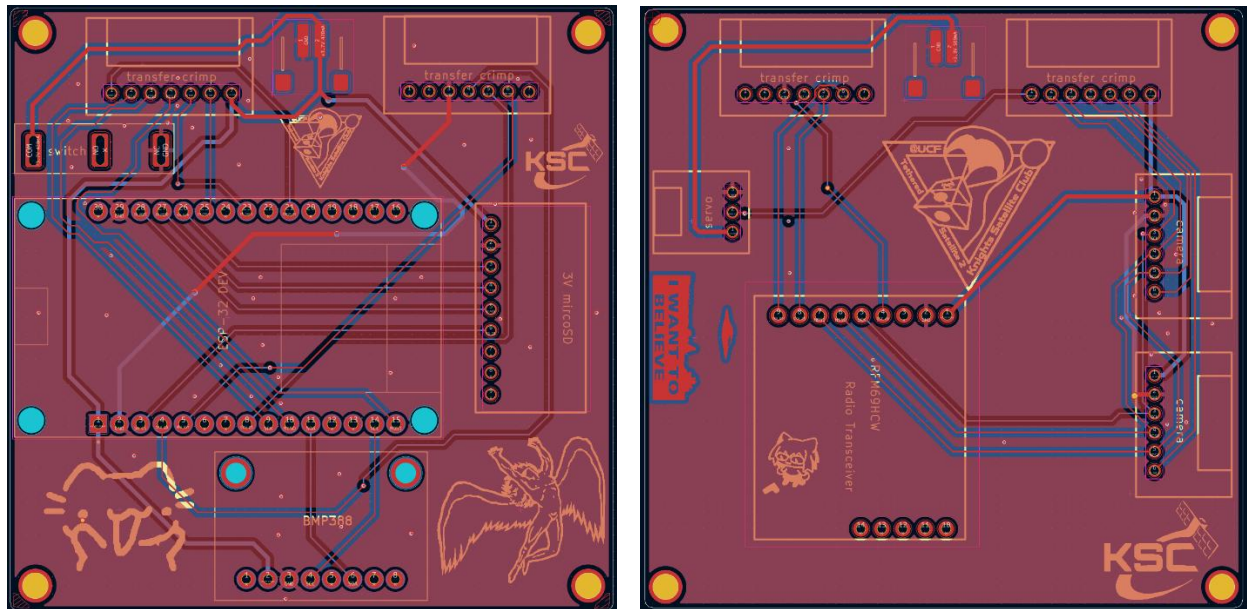


Figure 9: T-Sat 2A Electronics and Connection PCB's

Power System Development

The T-Sat 2A electrical system used two separate 3.7 V lithium-ion batteries. One battery was dedicated to the servo system and had a rated capacity of 500 mAh, while the second battery had a rated capacity of 420 mAh and powered the main electronics.

The decision to separate the power sources was intended to reduce the potential for servo operation to interfere with the main electronics. Servo actuation can introduce changes in the electrical load, making power-system behavior an important consideration when operating sensitive electronics such as cameras, sensors, and the ESP32.

Several power distribution configurations were considered and tested during development. The initial servo power configuration used a boost converter. Testing indicated that this configuration did not provide the required servo performance, leading the team to investigate an alternative power arrangement.

The final servo configuration used the dedicated 3.7 V battery without the boost converter. This change simplified the servo power circuit and removed an additional conversion component from the system.

A buck converter was also investigated for the main electronics power system. During testing, the team observed camera malfunctions and significant altitude measurement errors while using the converter. In response, the buck converter was removed from the final configuration, and the main electronics were connected directly to the battery system.

The observed behavior led to a revision of the power architecture. The final configuration reduced the number of power-conversion components and simplified the electrical distribution system. However, the specific electrical cause of the camera and altitude measurement problems should be documented separately if additional measurements or testing records are available.

Camera and Data Storage Integration

The imaging system consisted of two Arducam cameras connected to the ESP32. The cameras were intended to provide multiple fields of view while increasing the amount of visual data collected during the mission.

During initial development, the camera system and SD card module were configured to share an SPI bus. Testing revealed unreliable behavior, including image corruption, system crashes, and a configuration that operated successfully only once during testing.

Because image data must be transferred and written to the SD card, the storage process introduced additional demands on the shared communication interface. To improve system reliability, the SD card was moved to a separate SPI bus from the cameras.

The final software configuration assigned the cameras and LoRa module to the ESP32's primary SPI interface while the SD card was connected through a separate HSPI instance. The camera and radio continued to share their SPI bus, requiring software coordination to prevent simultaneous peripheral access.

The separation of the SD card from the camera SPI bus improved the observed system performance. However, the available testing did not conclusively establish that the shared SD card bus was the sole cause of the original failures. Other contributing factors may have included

chip-select behavior, peripheral initialization, software libraries, or signal integrity. Consequently, the improvement should be documented as an observed result of the architectural change rather than definitive proof of a single root cause.

The final software captured images from both cameras and stored the resulting JPEG files on the SD card. Camera data was written to separate image files using camera-specific filenames, allowing the images from each camera to be distinguished during post-processing.

Altitude Measurement System

Altitude measurements were obtained using the BMP388 barometric pressure sensor. The sensor was initialized through the I2C interface and configured using pressure oversampling, an IIR filter coefficient, and an output data rate.

The flight software calculated altitude from the measured atmospheric pressure using a standard barometric altitude equation. The calculation used a defined sea-level pressure reference, which was configured in the software as 1017.25 hPa.

Altitude data was stored locally on the SD card through the /Altitude.txt file. Each successful pressure measurement resulted in a calculated altitude value being appended to the file. This provided a local record of the altitude measurements rather than relying exclusively on radio transmission.

The system also transmitted altitude information to the ground station through the LoRa communication module. The transmission interval was controlled through a 25,000-millisecond timing variable, corresponding to a nominal interval of approximately 25 seconds.

Local storage was retained as an important part of the telemetry architecture because radio transmissions were not continuous. Recording the measurements on the CubeSat allowed altitude data to be preserved between transmissions and provided a means of reviewing the data after the mission.

LoRa Communication and Ground Station

The communication system used RFM95W LoRa radio modules operating at a configured frequency of 915 MHz. The flight computer used the onboard radio to transmit altitude information and receive commands from the ground station.

The ground station consisted of an ESP32, an RFM95W LoRa module, a user-operated button, and a laptop connected through USB power. The ground station software was developed using the Arduino IDE and provided telemetry through the serial monitor.

The communication system supported two primary operations: transmission of altitude data from the CubeSat and reception of the deployment command from the ground station. When the flight software received the Detach command, it set the corresponding control flags for the deployment sequence and prepared an acknowledgement message.

The deployment command provided the launch team with a means of initiating the parachute release sequence remotely. After receiving the command, the flight software sent an acknowledgement through the radio and subsequently activated the servo control function.

Because the LoRa radio and cameras shared the primary SPI bus, communication required coordination with camera operations. The software used a shared `spiBusy` flag to prevent radio processing while the camera system was actively capturing and saving images.

Servo and Deployment Control

The servo system was responsible for actuating the parachute release mechanism. The servo was connected to a dedicated control pin on the ESP32 and configured with an initial position of 0° and a deployment position of 90° .

When a valid deployment command was received, the software set the deployment control flags. The deployment function attached the servo, moved it to the initial position, and waited approximately three seconds before commanding the deployment position.

The three-second delay was implemented within the servo trigger function. This delay was separate from the time required for the main loop to reach the deployment function after receiving the radio command.

The servo system was powered using a dedicated 3.7 V, 500 mAh battery. This arrangement was selected following testing of the initial boost converter configuration and was intended to provide a simpler power path for the deployment actuator.

The deployment mechanism and its physical cutting-blade clearance are described in the mechanical design section. The electrical system provided the command processing and servo actuation required to operate that mechanism.

Flight Software Architecture and Timing

The flight software was developed to coordinate the camera system, altitude sensor, SD card, LoRa communication module, and servo. The software was structured around a continuous main loop that repeatedly checked for radio communication, performed an altitude measurement, captured and stored images, processed pending acknowledgement and deployment commands, and introduced a delay before beginning the next cycle.

The software used flags to coordinate tasks that could not safely operate at the same time. The `spiBusy` flag indicated when the camera system was using the shared SPI interface, preventing the LoRa processing function from accessing the bus during that period.

The camera capture process triggered both cameras and then saved the images to the SD card. The SD card was connected through a separate SPI bus, but the camera capture and image-saving operations were still treated as a protected period within the main loop.

The software also used the `servoActive` flag to prevent altitude transmission while the deployment sequence was active. A separate acknowledgement flag was used to schedule the transmission of the deployment acknowledgement.

Timing values were selected and adjusted during development to achieve more stable system operation. The main loop included a three-second delay, while altitude transmissions were controlled through a 25-second interval. These values affected the timing of communication, image processing, and deployment response.

The software's timing behavior was influenced by the duration of SD card writes, camera operations, sensor readings, and radio communication. Therefore, the actual time between tasks was not determined exclusively by the configured delay values.

Electrical Integration Issues and Solutions

The electrical development process involved several problems that required changes to the hardware architecture, PCB design, and software implementation.

The initial PCB development workflow in EasyEDA presented challenges involving component symbols and footprints. Transferring the design to KiCad provided greater control over custom components and supported the continued development of the PCBs.

Power-system testing identified problems with the initial boost converter configuration for the servo and the buck converter configuration for the main electronics. The final system removed these converters and used separate battery configurations for the servo and main electronics.

The camera and SD card SPI configuration also required revision. Initial testing of the shared bus configuration resulted in unreliable system behavior. Moving the SD card to a separate SPI bus improved the observed reliability, although the original failure mechanism was not conclusively isolated.

An additional issue occurred during testing of the pull-to-launch switch. Reversed crimp connections resulted in damage to the ESP32. Further testing also showed that the switch did not provide the desired true off state and could allow the battery to continue draining when the system was not actively operating.

The pull-to-launch switch was subsequently removed from the final configuration. Direct battery connection and disconnection were used for power control instead. This reduced the complexity of the activation system and avoided the specific issues encountered with the switch.

These experiences demonstrated the importance of testing individual electrical interfaces before completing full system integration. In particular, power supplies, connectors, SPI peripherals, and control switches should be verified independently before being incorporated into the complete CubeSat.

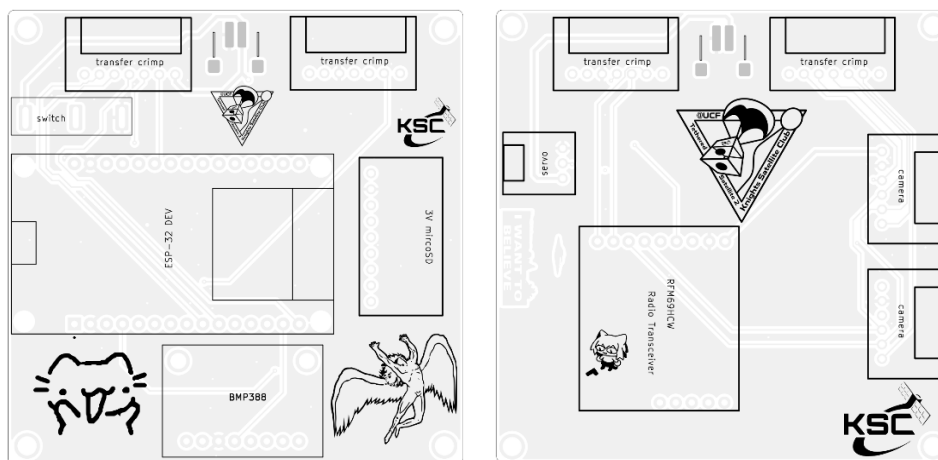


Figure 10: T-Sat 2A Electronics and Connection PCB's

Final Electrical Configuration

The final T-Sat 2A electrical configuration consisted of two custom PCBs, an ESP32 flight computer, two Arducam cameras, a BMP388 barometric altimeter, an RFM95W LoRa radio module, an SD card module, and a servo-controlled parachute release system.

The primary components and their associated functions were:

Component	Function
ESP32	Flight computer and subsystem coordination
Two Arducam cameras	Image acquisition
BMP388	Barometric altitude measurement
RFM95W LoRa	Wireless communication
SD card module	Local image and altitude data storage
Servo	Parachute release actuation
Two custom PCBs	Electrical distribution and subsystem interconnection
Two lithium-ion batteries	Separate power sources for the servo and main electronics

The final software configuration assigned the cameras and LoRa module to the primary SPI bus and the SD card module to a separate SPI bus. The system used standardized JST connectors for subsystem interconnection and direct battery-based power configurations following the converter-related testing.

The completed electrical system provided the interfaces required for imaging, altitude measurement, telemetry, data storage, and parachute deployment. Its final configuration was the result of multiple development iterations that addressed power-system performance, PCB development challenges, communication-bus reliability, and control-system integration.

The electrical design and flight software development process demonstrated the importance of testing hardware and software interfaces together. The final architecture reflected the team's efforts to simplify the system, improve reliability based on observed test results, and maintain a configuration that could be integrated into the mechanical housing and operated during the mission.

Flight Software and Ground Station

Software Architecture and Objectives

The T-Sat 2A software system was developed to support four primary mission objectives: image acquisition, altitude measurement, deployment control, and wireless communication. The system consists of two software components: the flight software running on the onboard ESP32 and the ground-station software used to monitor telemetry and initiate the parachute deployment sequence.

The onboard flight software manages the cameras, altimeter, SD card, LoRa radio, and deployment servo. It collects and stores mission data locally while transmitting selected information to the ground station. The ground station provides a user interface through a physical pushbutton and a LoRa communication link, allowing the operator to send a deployment command and receive altitude telemetry and deployment acknowledgements.

The complete software source code is available through the project's GitHub repositories. The final integrated flight-software version is maintained in Francesca's KSC repository:

- [Francesca's KSC Repository](#)

Earlier development code and PCB Hardware is available in Seth's repository:

- [Seth's T-SAT Repository](#)

During the initial development phase, the electronics system was not yet fully assembled because several components were still being delivered. As a result, the software was developed and tested using separate module-level programs. These programs allowed individual components, such as the cameras, altimeter, LoRa radio, and servo, to be tested independently before the hardware was fully integrated. Once the required electronics were available, the separate modules were combined into the integrated flight software.

Flight Software Functions

The flight software performs four main functions: image collection, altitude measurement, telemetry transmission, and deployment control.

Image Acquisition and Data Storage

The flight software controls two cameras, designated CAM1 and CAM2, to collect images during the mission. The images are stored on an onboard SD card for later analysis. The camera system and SD card use separate SPI buses to reduce communication conflicts and prevent the relatively slow SD-card writing process from interfering with camera operation.

The software captures images from both cameras during its main operating cycle. Based on the documented operating behavior, the system was designed to collect approximately five images per minute in total, equivalent to an image approximately every 12 seconds when considering the combined camera output. The exact image rate is dependent on image-capture and SD-card writing times and should therefore be reported as an observed performance value if it was measured during testing.

Local image storage provides a record of the mission even when wireless communication is unavailable or interrupted. This approach prevents the mission from depending on the LoRa link to transmit every image in real time.

```

183101.325 -> LoRa init SUCCESS
183101.325 -> Frequency: 915 MHz
183101.325 -> TX Power: 13 dBm
183101.325 ->
183101.325 -> === Ground Station Ready ===
183101.362 -> Press button to send 'Detach'
183101.362 ->
183146.130 -> Alt: -11.05 m
183146.130 -> <<< RSSI: -51
183218.926 -> Alt: -11.14 m
183218.926 -> <<< RSSI: -83
183257.886 -> Alt: -11.26 m
183257.886 -> <<< RSSI: -105
183334.728 -> Alt: -11.43 m
183334.728 -> <<< RSSI: -113
183411.877 -> Alt: -11.53 m
183411.877 -> Alt: -11.53 m
183411.877 -> <<< RSSI: -117
183448.533 -> Alt: -11.67 m
183448.533 -> <<< RSSI: -119
183525.593 -> Alt: -11.73 m
183525.593 -> <<< RSSI: -124
183639.991 -> Alt: -11.86 m
183639.991 -> <<< RSSI: -129
183717.971 -> Alt: -11.99 m
183717.971 -> <<< RSSI: -128
183755.948 -> Alt: -12.08 m
183755.948 -> <<< RSSI: -124
183717.971 -> Alt: -11.99 m
183717.971 -> <<< RSSI: -128
183755.948 -> Alt: -12.08 m
183755.948 -> <<< RSSI: -124
183834.242 -> Alt: -12.19 m
183834.242 -> <<< RSSI: -126
183846.186 -> Sending 'Detach' message...
183847.266 -> Message sent!
183847.750 -> Waiting for reply...
183912.560 -> (ignored non-ACK packet)
183912.737 -> No reply received 1
184108.821 -> Alt: -12.56 m
184108.821 -> <<< RSSI: -130
184147.243 -> Alt: -12.62 m
184147.243 -> <<< RSSI: -127
184226.192 -> Alt: -12.70 m
184226.192 -> <<< RSSI: -114
184147.243 -> Alt: -12.62 m
184147.243 -> <<< RSSI: -127

```

Figure 11: Running Serial Monitor Status

Altitude Measurement and Logging

Altitude data is collected using the onboard barometric altimeter. The measured altitude is transmitted to the ground station through the LoRa communication link and is also recorded on the SD card.

The software prioritizes local altitude logging so that altitude information can continue to be recorded even when radio transmissions are delayed or unavailable. The documented radio transmission interval is approximately 25 seconds. This interval applies to the transmission of altitude telemetry and should not be confused with the frequency at which altitude data is locally recorded.

LoRa Communication

The T-Sat 2A communication system uses a LoRa radio module to exchange information between the flight computer and the ground station. The radio supports both transmission and reception, allowing the flight system to send telemetry and receive commands from the ground.

The LoRa radio shares an SPI bus with the cameras. Because simultaneous access to the SPI bus can cause communication conflicts, the flight software uses an `spiBusy` flag to coordinate access. When the cameras are using the SPI bus, LoRa communication is temporarily

prevented. Once the camera operation is complete and the bus is available, the software can resume radio communication.

This coordination is important because an incoming deployment command or acknowledgement could otherwise interrupt a camera transaction or cause the flight software to become unresponsive.

The flight software also uses a `servoActive` flag to manage communication during the deployment sequence. While the servo is active, normal altitude transmissions are suspended. This allows the software to prioritize the deployment process and receive or transmit the relevant deployment acknowledgement.

Deployment Command and Servo Control

The ground station initiates the deployment sequence when the operator presses the physical pushbutton. The ground-station software detects the button's rising edge and sends a LoRa message containing the command `Detach`.

After transmitting the command, the ground station waits for an acknowledgement from the flight system. The expected acknowledgement is the message `Detached`. The ground station waits for up to 25 seconds for this response because the flight software may require time to process the command and operate the servo.

The ground station does not immediately accept other deployment commands while it is waiting for a response. A 30-second cooldown period is also implemented after a successful acknowledgement to reduce the possibility of repeated deployment commands.

The acknowledgement confirms that the flight software processed the command and returned the expected response. It does not independently verify that the physical parachute-release mechanism successfully deployed unless a separate mechanical sensor or visual verification system is included.

Ground Station Software

The ground station is implemented using an ESP32 and an RFM95 LoRa radio module. The radio is initialized using the RadioHead `RH_RF95` library and configured to operate at 915 MHz with a transmit power of 13 dBm.

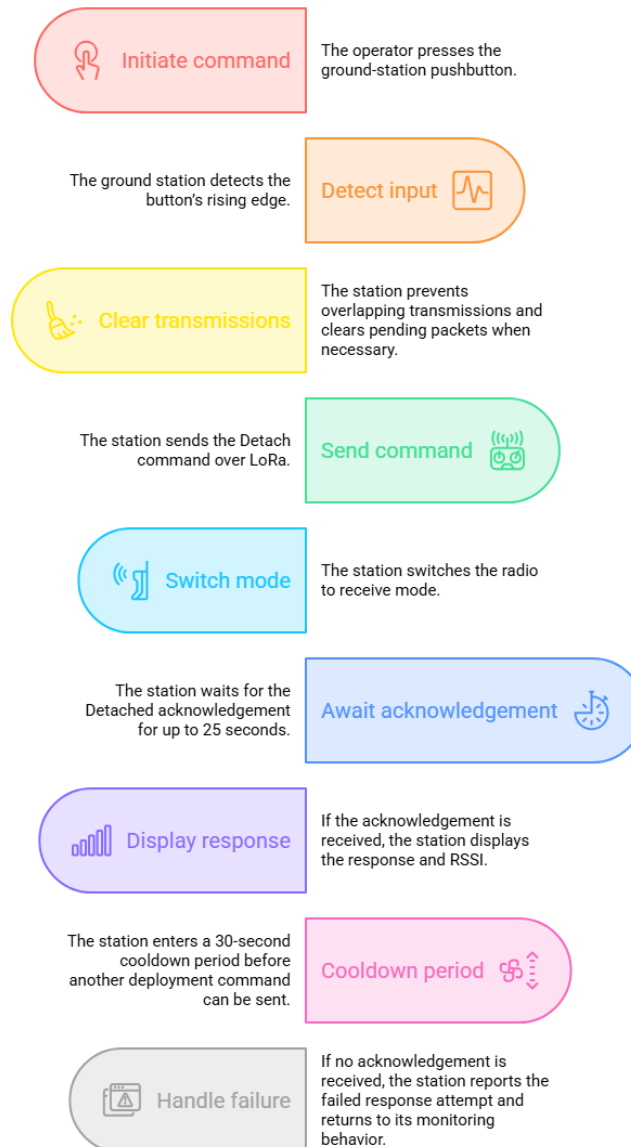
During initialization, the software:

1. Starts serial communication at 74,880 baud.
2. Initializes the SPI interface.
3. Resets the LoRa radio module.
4. Configures the deployment pushbutton as an input.
5. Initializes the LoRa radio.
6. Sets the operating frequency to 915 MHz.
7. Sets the radio transmit power to 13 dBm.

Once initialization is complete, the ground station enters its primary operating loop. It continuously checks for incoming packets and displays received altitude messages on the serial monitor. Packets beginning with Alt: are treated as altitude telemetry, while other packets are ignored unless they are being processed as part of the deployment acknowledgement sequence.

The ground station also records the received signal strength indicator (RSSI) for incoming packets. RSSI provides an indication of received signal strength and can be used during testing to evaluate the communication link.

Deployment Communication Sequence



The ground station uses packet flushing and short delays to reduce the likelihood that previously received altitude packets will interfere with the deployment command or acknowledgement. These measures are intended to separate routine telemetry traffic from the higher-priority deployment communication.

Timing and Communication Considerations

The software timing was selected to balance data collection, radio communication, and deployment reliability. The flight system must perform camera operations and SD-card writes while also maintaining communication with the ground station. Because these activities share hardware resources, the software uses timing delays and status flags to reduce interference.

The documented main-loop delay is approximately three seconds. Testing indicated that shorter delays produced less reliable radio behavior, while longer delays reduced timing accuracy. The three-second delay was therefore selected as the most stable configuration during development.

The ground station uses a 200-millisecond waiting period between packet activity and deployment transmission to reduce the possibility of packet overlap. It also waits approximately 500 milliseconds after sending the deployment command before beginning its acknowledgement monitoring sequence.

The software timing values represent programmed delays and communication intervals. They should be distinguished from measured end-to-end response times, which include camera processing, SD-card writing, radio transmission, and flight-computer processing.

Software Development and Verification

Software development began with independent tests of individual hardware components. This approach allowed the team to validate component operation before full system integration and reduced the difficulty of diagnosing problems in the complete system.

As the electronics became available, the individual modules were integrated into the flight software. Integration required coordination between camera capture, SD-card storage, altitude sensing, LoRa communication, and servo control. Particular attention was given to SPI-bus conflicts because the cameras and LoRa radio shared one SPI interface.

The final software configuration incorporated communication-control flags, deployment-state handling, local data logging, and ground-station acknowledgement monitoring. The source code and earlier development modules are retained in the project's GitHub repositories to document the software development process and support future troubleshooting or reproduction of the system.

Budget and Resources

Part Name/Item Name	Date Issued	Part Number	Description	Quantity Requested	Unit Price (USD)	Subtotal (USD)	Supplier / Vendor	Estimated Delivery Date from Date Issued	Status	Notes / Comments
3M Nylon Parachute	10/1/2025	K2085	3M Nylon Parachute	2	\$18.89	\$37.78	Amazon	10/06/2017	Received	Price increase in price shipping and taxes \$7.87 + \$1.82
Black PETG Filament	10/1/2025	ASIN: B08WCPYH1	3DPrinter.com PETG Filament	2	\$9.99	\$19.98	Amazon	3-Oct	Received	Only ordering one roll now for test prints possibility new roll needed for final assembly
TI/AD9833 3.3V Buck Converter	10/30/2025	Product ID: 47111	Buck Converter	3	\$3.95	\$11.85	adafruit			
WPM0500 Transceiver Radio	10/30/2025	Product ID: 3079	SDR	2	\$9.99	\$19.98	adafruit			
CH3225 250k Ohm	10/30/2025	408-88154-MS07	SDR	2	\$19.99	\$39.98	Amazon			
Spring Antenna 915MHz	1/23/2026	Product ID: 42889	Spring Antenna	10	\$9.99	\$99.90	adafruit		Received	
Module 861	10/30/2025		Module 861	1	\$4.99	\$4.99	Amazon			
Adafruit BMP388 Altitude Module	10/1/2025	Product ID: 3996	Adafruit BMP388 Altitude - STEMMADOT	3	\$9.99	\$29.97	adafruit	USPS - USPS Ground Advantage - 5 Day(ri)	Received	
420mAh Li-Ion Battery	1/23/2026	Product ID: 4276	Li-Ion Battery - 420mAh	2	\$9.99	\$19.98	adafruit	USPS - USPS Ground Advantage - 5 Day(ri)	Received	
500mAh Li-Ion Battery	1/23/2026		Li-Ion Battery - 500mAh	2	\$7.99	\$15.98	adafruit	USPS - USPS Ground Advantage - 5 Day(ri)	Received	have chargers in storage plus get extra for test A and B project
ALPCB Water	2/12/2026	N/A	ALPCB Water	2	\$96.30	\$192.60	ALPCB	Arrived 2/26/2026	Received	
3 Pin ESP32 Dev Boards	10/1/2025	ASIN: B08WCPYH1	3 Pin ESP32 Dev Boards	1	\$26.99	\$26.99	Amazon	3-Oct	Received	10 Pack Spite With T-Sat 2B
ArduCam Mega 20 Camera	10/30/2025	ASIN: B08WCPYH1	ArduCam Mega 20 Camera	2	\$19.99	\$39.98	Amazon	Thursday, Oct 9 - Sunday, Oct 12	Received	10 Pack Spite With T-Sat 2B
16GB SD Card	10/30/2025	Product ID: 4614	SD Card	5	\$3.99	\$19.95	adafruit			
10 Pin 1A 10001003	10/30/2025	Product ID: 1389	10 Pin 1A 10001003	5	\$1.99	\$9.95	adafruit			
10 Pin 2 Pin 1A 10001003	10/30/2025	Product ID: 4682	10 Pin 2 Pin 1A 10001003	3	\$3.99	\$11.97	adafruit		Received	replaced previous model for reliability and documentation use of the module
Micro SD SPI Breakout Board	1/23/2026	Product ID: 4682	Micro SD SPI Breakout Board	3	\$3.99	\$11.97	adafruit		Received	heavily affected highly since calculating every single one with their own shipping fee
					Total:	\$389.24				

Figure 12: T-Sat 2A Budget

The T-Sat 2A project maintained a separate project-level budget to track the materials and components purchased specifically for the development and testing of the CubeSat. Expenses associated with launch operations, shared wiring, and other resources used across the Knight's Satellite Club were not included in this total, as those expenses were managed through the broader club budget.

The recorded project purchases totaled **\$389.24**. The largest portions of the budget were associated with the electrical and avionics system, including the ESP32 development boards, cameras, LoRa radio modules, BMP388 altitude sensors, SD card hardware, batteries, connectors, and custom PCBs. Mechanical and deployment hardware was also purchased, including PETG filament, the nylon parachute, servo, antenna, and cutting blade.

Several components were purchased in quantities greater than what was ultimately required for a single T-Sat 2A vehicle. This was intentional in cases where components were purchased as multipacks or where additional hardware could be shared with other Knight's Satellite Club projects. For example, the ESP32 boards, cameras, and SD cards were purchased in quantities that could be distributed between T-Sat 2A and T-Sat 2B. Additional batteries were also retained with the club's existing chargers and storage inventory for use by T-Sat 2A, T-Sat 2B, and future testing.

The budget also includes components that were investigated or used during development but were not incorporated into the final flight configuration. Examples include power-conversion hardware that was evaluated during electrical testing. These purchases are retained in the project budget because they represent actual development costs, even though the components did not ultimately fly. Accounting for these items provides a more accurate representation of the resources required to develop the final system and demonstrates that not every component purchased during an engineering project necessarily becomes part of the final vehicle.

The budget was also affected by shipping and tax costs, particularly for individual orders where purchasing components separately resulted in additional delivery charges. These costs were included in the recorded purchase totals rather than being excluded from the component cost.

Overall, the project budget supported the development, testing, and final integration of the T-Sat 2A mechanical and electrical systems while also providing additional components for future iterations and related KSC projects. The complete itemized budget, including quantities, unit prices, suppliers, delivery information, and notes regarding component usage, is provided in the accompanying budget table.

Conclusion

The development of T-Sat 2A resulted in a completed 2U tethered satellite system that brought together mechanical, electrical, software, communications, imaging, altitude measurement, and parachute-deployment systems into one integrated vehicle. Throughout the project, the design continued to evolve as the team encountered new challenges, tested different approaches, and learned from both successful and unsuccessful results.

Not every idea or component made it to the final configuration, but each iteration contributed to a better understanding of the system and helped guide the decisions that ultimately produced the completed vehicle. From redesigning electrical systems and resolving communication issues to refining the mechanical structure and testing the deployment system, the project provided valuable experience that extends beyond the final hardware itself.

T-Sat 2A represents the combined effort of the students who designed, tested, built, and supported the project. While there were failures, unexpected problems, and decisions that could be improved in future iterations, those experiences are an important part of the engineering process. The knowledge gained throughout the project provides a foundation for future members of the Knight's Satellite Club to build upon, improve, and continue developing the T-Sat program.

Ultimately, T-Sat 2A became more than a collection of individual subsystems. It became a completed engineering effort shaped by the challenges encountered along the way, and the lessons learned from those challenges will remain one of the project's most valuable results.



Figure 13: Final Assembly of T-Sat 2A

References

1. *NASA Systems Engineering Handbook*. NASA/SP-2016-6105 Rev 2, National Aeronautics and Space Administration, Washington, D.C., 2016. Available at: <https://www.nasa.gov/reference/systems-engineering-handbook/>
2. *NASA Procedural Requirements 7123.1C – Systems Engineering Processes and Requirements*. National Aeronautics and Space Administration, Office of the Chief Engineer, Washington, D.C. Accessed via NASA Directives and Regulations System (NODIS): https://nodis3.gsfc.nasa.gov/displayCA.cfm?Internal_ID=N_PR_7123_0001_&page_name=AppendixG
3. *Knight's Satellite Club – UCF Wiki*. University of Central Florida, Knight Space Club, Orlando, FL. Available at: <https://wiki.kscucf.org/>
4. *NASA Technical Memorandum 20110008314 – Ares V Lunar Capabilities Concept Review*. National Aeronautics and Space Administration, Marshall Space Flight Center, Huntsville, AL, 2011. (Uploaded reference document used for format and structure guidance.)
5. *SEDS Project 39-B Critical Design Review – Final Draft v2*. Students for the Exploration and Development of Space (SEDS), University of Central Florida, 2022. (Uploaded reference document used for subsystem documentation style and presentation layout.)
6. *T-Sat-2 Systems Engineering Handbook – Draft Template*. Knights Satellite Club, University of Central Florida, 2025. (Internal document prepared by A. Salazar; used for subsystem formatting and requirement structure.)
7. *T-Sat-2 Project Timeline*. Knights Satellite Club, University of Central Florida, 2025. (Internal scheduling reference for project phase mapping and milestone planning.)
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